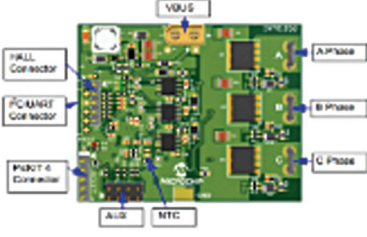
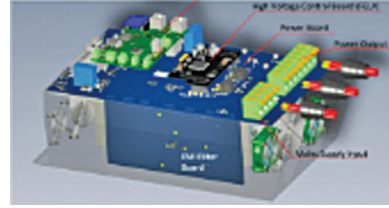


The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world

For Different Applications

Low Powered E-Scooter Motor Drive	3-Phase PMSM Drive	22kW Industrial Motor Drive
		
BLDC	PMSM	3-phase induction motor
18V to 24V	230V AC	400V AC
350W	1.5kW	22kW
Reference design for e-bike motor controller that can take input from 18V to 24V and deliver a maximum power of 350W. The design is capable of driving both a BLDC motor as well as a PMSM motor of up to 350W power.	This reference design for a 3-phase PMSM drive with a 32-bit microcontroller includes a low-voltage MOSFETs power stage and a communication stage. The design is suitable for motor drives for drones, compressors, air-conditioners, fans, air extractors, pumps, and industrial drives, etc.	This reference design is for a 22kW industrial motor drive for 3-phase 400V AC grids. The motor driver can take input from 380 to 480V AC. The reference design is highly customisable and can be used for both low-powered and high-powered applications.
The dsPIC33CK64MP105 DSC motor controller from Microchip is a high-efficiency design suitable for e-mobility solutions. It is suitable for low-powered e-scooter and e-bikes. The reference design can take input from 18V to 24V and deliver a maximum power of 350W. It is capable of driving both a BLDC motor as well as a PMSM motor of up to 350W power. The brain of this design is digital signal controller (DSC), dsPIC33CK. The single-core dsPIC33CK DSCs accelerate digital signal processor (DSP) performance for time-critical applications. This design also employs six MOSFETs with low RDS(on), which enhances safety as the MOSFETs can handle high phase currents without excessive conduction losses. The design offers high efficiency, compact dimensions, and low cost while providing high performance.	The YROTATE-IT-RX111 is a motor drive for a 3-phase permanent magnet synchronous motor (PMSM) or AC brushless motor that employs an advanced sensorless field oriented control algorithm. The design is compatible with other existing motor control reference platforms with external power stages delivering up to 1.5kW at 230V AC. The reference design features a microcontroller (MCU) from Renesas 32-bit RX MCU family. The main advantage of the design is that it is simple and economical. The reference design allows the user to monitor and tune various parameters using the graphical user interface provided by the company. The design is highly customisable and allows the user to select the best switching frequency and control frequency to adapt the control dynamics suitable to the application requirements. This enables any designer to make the right compromise between system dynamics, CPU load, and overall efficiency.	The REF-22K-GPD-INV-EASY3B is a reference design of an industrial-grade motor drive from Infineon. The motor drive is suitable for 3-phase 400V AC grids and has a nominal power of 22kW. It includes a power module, current sensor, gate driver IC, and a 1.7kV SiC-MOSFET. The reference design features two microcontrollers and can be divided into five separate boards: power board, EMI filter, high-voltage logic board, low-voltage interface board, and DC-link board. This makes the reference design highly customisable with respect to specification and size. Furthermore, it allows for better isolation between different sub-systems, making the reference design safer and more reliable. The design also features software that enables full control and communication via touch screen and PC-GUI. The reference design can take 3 ϕ input of 380 to 480V and provide a wide range of output from 0 to 456V with a rated current of up to 45A. The reference design can be used for both low-powered motor control as well as high-power applications, such as conveyor belts and compressors.
The reference design is suitable for use in e-bikes, electric lawnmowers, etc.	The reference design can be used to make motor drives for drones, compressors, air-conditioners, fans, air extractors, pumps, and industrial drives, etc.	The reference design can be used for both low-powered motors like refrigerator compressors and high-powered motors for conveyor belts, and running large compressors for warehouse chillers.
Microchip	Renesas	Infineon
https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-low-powered-e-scooter-motor-drive	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-3-phase-pmsm-drive	https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-an-industrial-motor-drive